

Spectrum Sensing for Symmetric α -Stable Noise Model With Convolutional Neural Networks

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Abstract—The key role of spectrum sensing in cognitive radios attracted substantial research attention to improve the performance of detectors. We consider a general model for the receiver noise with the potential of generalization towards modeling noise in various environments. This general noise model describes accurately noise characteristics ranging from the Gaussian noise to the severe impulsive noise. However, many previous studies are based on the ideal Gaussian noise, and they cannot capture the non-Gaussian models. To provide a robust detector against different behaviors of the noise in various environments, we employ a convolutional neural network (CNN) compatible with different noise models. The proposed CNN detector is data-driven, and due to its single-dimensional input layer, it is consistent with the received signal and requires no pre-processing. The likelihood ratio test (LRT), the Wald, and the Rao tests for this problem are derived to enrich the paper with comparative evaluations of the proposed CNN and conventional model-based approaches and other neural networks. Although various simulated scenarios substantiate the general superiority and robustness of the CNN-based against impulsive noise and mismatch of parameters, it requires higher computational complexity than other discussed detectors.

Index Terms—Cognitive radio, spectrum sensing, deep learning, neural networks, CNN, α -stable, impulsive noise.

I. INTRODUCTION

RISE of data transmission through wireless channels necessitates intelligent communicating systems capable of using the spectrum opportunistically. Cognitive radios (CRs) as a novel resolution have been proposed based on the software-defined radios by researchers to bring efficiency to the utilization of the RF bands [1]. In a cognitive network (CN), a primary user (PU) is free to use its allocated frequency bands whenever it desires to transmit data. Concurrently, the network permits secondary users (SUs) to exploit the bands which are not in the service of the PUs. When the PU starts transmitting its data, the SU has to vacate its channels immediately to avoid interfering with the PU's signal.

Accordingly, the process of identifying an idle or active frequency channel by the SU, called spectrum sensing, plays an essential role in CNs. The SU has to sense the spectrum,

and then, it decides about the state of the channel. Idle and active bands must be detected accurately to raise the efficiency of the CN without causing interference for PUs' communication. On the other hand, the detection process has to be robust for a variety of real-world conditions and find idle bands quickly to provide more time for the opportunistic use of the SU.

Numerous research and papers take into account the different aspects of the spectrum sensing problem to achieve higher detection performance. Highly informative surveys on spectrum sensing in [2], [3] investigate various aspects of the problem. The spectrum sensing can be formulated as a simple hypothesis testing problem and addressed by a mathematical lemma leading to the optimum solution called the Neyman-Pearson (NP) lemma. However, for composite hypothesis testing problems with unknown parameters in the hypotheses, the NP detector cannot be found. Thus, other methods, namely LRT, the Wald test, and the Rao test can be employed, instead [4], [5]. Due to various assumptions on the model and parameters of the hypotheses, numerous detectors and tests are proposed in the literature. Estimator-correlator (E-C) is an optimum detector based on the NP lemma for differentiating a random signal with a known covariance matrix from the Gaussian noise with a known variance. For white signals, the E-C reduces to the well-known energy detector (ED) [5], [6]. Mostly, the transmitted signal by PUs and its covariance are not available in the SUs' receiver; consequently, E-C is not a feasible solution. In this paper, where we assume the transmitted signal is white, E-C is equivalent to ED. Thus, we considered its performance as a benchmark for performance evaluations. Additionally, many papers studied detectors based on the eigenvalues of the sample covariance matrix (SCM) [7]–[10] for multi-antenna SUs, and some studies employed cyclostationary features for spectrum sensing [11], [12].

In addition to detection theory techniques, plenty of machine learning methods are also employed for spectrum sensing [13]. Excellent performance of deep neural networks (DNNs) has led to employing them in a wide variety of applications [14]–[17]. CNNs and deep belief networks (DBN) in image classification, pattern recognition, and object detection [18], [19], recurrent neural networks (RNN) for speech recognition [20], generative adversarial networks (GAN), and variational autoencoders (VAE) for pattern generation and data augmentation [21] all have contributed to improving results. Thus, promising spectrum sensing approaches based on deep learning are also expected to be realized.

Furthermore, DNNs are able to extract information from training data through the learning phase, which makes them

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adaptable to different models. In other words, detectors provided by DNNs are data-driven, enabling them to generalize the DNN-based detectors for other signal models and conditions. To exploit this advantage of DNNs in signal processing, we consider a comprehensive model for noise distribution, symmetrical α -stable ($S\alpha S$) distribution, which is able to explain the behavior of the noise signal under various situations. To propose a robust solution, our model has to capture various environmental noise effects. The most widely used distribution to model the noise signals is the Gaussian distribution. This is partly due to tractability of the analysis when Gaussian noise model is used. This model, however, cannot properly present diverse phenomena with impulsive behavior generated by either natural or man-made sources. Examples of such phenomena include atmospheric effects such as lightening, electrical arcs in power supplies, electrical motors, car ignition, and fluorescent lights. These are routine phenomena occurring in everyday life which are generally ignored in the models, while their effects on the performance of the systems can be significant. These noises are categorized as heavy-tailed noises. As a consequence, the general statistical model introduced by $S\alpha S$ distribution can also incorporate impulsive noise with heavy-tailed distribution, and it is more compatible with empirical data [22]–[24]. In the CN context, authors of [24] verified $S\alpha S$ model with empirical data. Then, for multi-antenna receiver, they suggested a detector based on covariation and fractional lower order moments (FLOMs) and eigenvalues of SCM after Myriad filters, which suppress impulsive noise. A method based on combining order statistics was developed and discussed in [25]. Detectors employing kernels and FLOMs were studied under impulsive noise in [26]–[29].

Moreover, several works examined the spectrum sensing by DNNs under the Gaussian noise contamination. In [30], an artificial neural network (ANN) was proposed whose input was the power of signals. In [31], energy and cyclostationary properties of signals were extracted and employed for CNN with a single convolution layer to detect the PU presence. Authors of [32] suggested a new cooperative spectrum sensing based on a CNN with the energy statistic of all the sensing nodes. Spectrum sensing for PUs with orthogonal frequency-division multiplexing (OFDM) by feeding spectral correlation density (SCD) into a CNN was studied in [33], [34]. Signal detection by inserting SCM of the received signal from multiple antennas into the CNN as an image was proposed by [35]–[37]. A network including stacked autoencoders (SAE) with logistic regression is developed and employed for the detection of OFDM symbols in [38]. Detailed properties of detectors proposed by these works are presented in Table II.

Basically, unlike the Gaussian noise, spectrum sensing with SCM or energy features for stable noise models due to undefined second-order moments encounters critical issues. In the following, we show performance degradation of ED for the noise signal with heavy-tailed distributions. Moreover, we propose a method suitable for SUs with single antenna based on CNN which is also robust against impulsive noise. Here, we briefly list the contributions of this paper:

- Firstly, the paper formulates the complex signal model and spectrum sensing problem with complex $S\alpha S$ noise distribution with independent real and imaginary parts, while by this assumption, we lose circular symmetry of the noise except for the case of the Gaussian noise distribution [39].
- Next, the LRT, the Rao test, and the Wald test, model-based detectors, for the presented signal model in this paper are derived for different cases of the stable noise model. To the best of our knowledge, it is the first time that the LRT and the Rao tests are derived for the complex Cauchy distribution without circular symmetry.
- We propose a CNN detector for the general case of the stable noise model, benefiting from its data-driven detection performance, which is not model-dependent in contrast to other discussed methods. Despite the majority of the works employing CNN in spectrum sensing, the input layer of the proposed CNN is a one-dimensional array composed of the original received samples. As a result, the trained network provides a robust detector against not only these models' variations but also the mismatch of parameters and discrepancies in train and test datasets. Moreover, we suggest a novel method for having desired false alarm probability with the proposed CNN detector according to the bias of the last neuron of the last fully connected layer.
- Next, for qualitative comparisons, Table II incorporating several model-based and also data-driven proposed detectors in the literature with their properties is presented and discussed. Furthermore, extensive simulations between detectors, which can handle impulsive noise and are compatible with our signal model, demonstrate the robustness of the proposed detector against the impulsive noise models, and its better detection performance for all the discussed cases of heavy-tailed noise. Moreover, for the Gaussian distribution, ED is optimum to detect unstructured signal [5], [6], and the proposed CNN's performance is close to that of ED. However, as will be discussed in the following, the proposed detector requires higher computational complexity due to its deep convolutional feature extracting part. Also, discussions on reducing the complexity of the proposed CNN without degrading its performance are presented at the end.

The rest of the paper comes as follows: In Sections II, signal model and noise model are introduced and explained. Spectrum sensing problem and related conventional solutions and the proposed DNN for the problem are discussed in Section III. Simulation results and discussions are presented in Section IV. Finally, the reader can find the conclusion of the paper in Section V.

II. SYSTEM MODEL

Here, we regard a basic single-input single-output (SISO) signal model, easy to be extended into the multiple-antennas models. Each PU and each SU has a single antenna for transmitting and receiving signal through a flat fading channel. Also, the fading coefficient of the channel is modeled as slow fading in short sensing intervals [7]–[10]. In the PU receiver,

after in-phase (I) and quadrature (Q) downconversion and analog to digital (A/D) conversion, the sample of the baseband equivalent signal is

$$r[n] = hs[n] + w[n], \quad (1)$$

for $n = 1, \dots, N$, where N is all samples received in one sensing interval. $s[n]$ and $r[n]$ are the n -th sample of the transmitted signal by PU, and the received signal in the SU. The channel coefficient is denoted by h . In this paper, we consider a general case where the noise samples are independent and identically distributed (i.i.d.). Each I and Q components of the noise sample are denoted respectively by $w_I[n]$ and $w_Q[n]$, and they are related to $w[n]$ by $w[n] = w_I[n] + jw_Q[n]$. Both of them follow $S\alpha S$ distribution, $w_I[n], w_Q[n] \sim S\alpha S(\mu, \alpha, \gamma)$, which is a subclass of stable distribution and has no closed-form probability density function (PDF). The characteristic function of a random variable from α -stable distribution is [24], [39]

$$\Phi(\psi) = \exp\{j\mu\psi - \gamma|\psi|^\alpha[1 + j\beta\text{sign}(\psi)\kappa(\psi, \alpha)]\}, \quad (2)$$

where

$$\kappa(\psi, \alpha) = \begin{cases} \tan\frac{\pi\alpha}{2}, & \alpha \neq 1 \\ \frac{2}{\pi}\log|\psi|, & \alpha = 1, \end{cases} \quad 0 < \alpha \leq 2, \quad -1 \leq \beta \leq 1, \quad 0 < \gamma, \quad (3)$$

and $\text{sign}(\cdot)$ is sign function. When β is equal to zero, α -stable distribution becomes symmetric around the location parameter, μ_0 , and reduces to $S\alpha S$ subclass. Consequently, by plugging $\beta = 0$ into (2), the characteristic function of $S\alpha S$ distribution can be found by

$$\Phi(\psi) = \exp\{j\mu_0\psi - \gamma|\psi|^\alpha\}. \quad (4)$$

The characteristic exponent, α , is a key parameter to change the heaviness of the tails of the PDF around the location parameter, μ_0 . The dispersion parameter, γ , is related to the spread and deviation of the PDF. It is worthwhile to take into consideration two special cases of the $S\alpha S$ distribution: (i) $\alpha = 1$, the Cauchy distribution, which statistically models the severe impulsive noise with the heaviest tails; (ii) $\alpha = 2$, the conventional Gaussian noise with normal distribution.

For better signal detection, it is of great importance to employ an accurate distribution for the noise signal. $S\alpha S$ distribution in comparison with normal distribution has one extra parameter to describe the noise process more accurately. This extra parameter, α , enables the model to match noise measurements of different physical environments in which the noise signal may not follow a normal distribution. As described earlier, the SU has to sense the spectrum while it is influenced by any of the enumerated noises. To achieve acceptable performance in various environments, it is vital to use a detector compatible with all environments with different noise models. In other words, for a vast range of noise contamination, modeled by $1 \leq \alpha \leq 2$ (from impulsive noise distributed by the Cauchy distribution model to the conventional Gaussian noise), a robust detector is expected to perform appropriately in all these different environments.

III. SPECTRUM SENSING PROBLEM

In the rest of the paper, we investigate the spectrum sensing problem for the aforementioned general signal model with two dissimilar detection theory methods and DNNs.

A. Binary Hypothesis Testing Problem With LRT

The presence of a PU, alternative hypothesis, H_1 , and an idle channel, null hypothesis, H_0 , can form a binary hypothesis testing problem given by

$$\begin{cases} H_0 : r[n] = w[n] \\ H_1 : r[n] = hs[n] + w[n] \end{cases} \quad (5)$$

for $n = 1, \dots, N$. The distribution of the received signal conditioned on both hypotheses is

$$\begin{aligned} r_I[n]|H_0 &\sim S\alpha S(0, \alpha, \gamma), \\ r_Q[n]|H_0 &\sim S\alpha S(0, \alpha, \gamma), \\ r_I[n]|H_1 &\sim S\alpha S(\mu_I[n], \alpha, \gamma), \\ r_Q[n]|H_1 &\sim S\alpha S(\mu_Q[n], \alpha, \gamma), \end{aligned} \quad (6)$$

where

$$\begin{aligned} r[n] &= r_I[n] + jr_Q[n], \\ s[n] &= s_I[n] + js_Q[n], \\ \mu_I[n] &= \text{Re}\{h\}s_I[n] - \text{Im}\{h\}s_Q[n], \\ \mu_Q[n] &= \text{Re}\{h\}s_Q[n] + \text{Im}\{h\}s_I[n]. \end{aligned} \quad (7)$$

$\text{Re}\{\cdot\}$ and $\text{Im}\{\cdot\}$ indicate real and imaginary parts of a complex number. It is assumed that the location parameter for both I and Q components of noise is zero, and γ and α are available for the SU, but it is unaware of h , and $s[n]$. Thus, we assume h , and $s[n]$ are deterministic unknown parameters. The solution for the problem in (5) is a function of the received signal with the form of

$$\begin{matrix} H_1 \\ f(\mathbf{r}) \gtrless \eta, \\ H_0 \end{matrix} \quad (8)$$

where $\mathbf{r} = [r[1] \dots r[N]]^T$ is a vector with size of $N \times 1$. The test statistic, $f(\mathbf{r})$, is compared to the detection threshold, η , and then one of the hypotheses is confirmed for the state of the sensed spectrum.

The desired solution for a binary hypothesis testing problem is expected to obtain efficient statistics or metrics, such as the maximum probability of detection, P_d , or true positive, and the minimum false alarm probability, P_{fa} , or false positive. For both these probabilities, we have $P_d = \Pr\{f(\mathbf{r}) > \eta|H_1\}$, $P_{fa} = \Pr\{f(\mathbf{r}) > \eta|H_0\}$, where $\Pr\{\cdot\}$ shows the probability of an event. To solve the binary hypothesis testing problems with unknown parameters in (5), we need to define the test vector as $\boldsymbol{\theta} = [\boldsymbol{\mu}_I \ \boldsymbol{\mu}_Q]^T$ (see (30)). Consequently, by defining $\boldsymbol{\mu}_I = [\mu_I[1] \dots \mu_I[N]]^T$, and $\boldsymbol{\mu}_Q = [\mu_Q[1] \dots \mu_Q[N]]^T$, under H_0 , the amount of the test vector is known and equals to $\boldsymbol{\theta}_0 = [\boldsymbol{\mu}_I \ \boldsymbol{\mu}_Q]^T = [\mathbf{0} \ \mathbf{0}]^T$, while for H_1 , it is $\boldsymbol{\theta}_1 = [\boldsymbol{\mu}_I \ \boldsymbol{\mu}_Q]^T \neq \boldsymbol{\theta}_0$. Thus, the test vector under H_1 is required to be estimated by the maximum

likelihood estimation (MLE) and is denoted by θ_1^* . Now, the conventional LRT can be derived by [4], [5]

$$f(\mathbf{r}) = \max_{\theta_1} L(\mathbf{r}|\theta_1) - L(\mathbf{r}|\theta_0). \quad (9)$$

The log-likelihood function (LF) parametrized by the vector test under H_j is denoted by $L(\mathbf{r}|\theta_j)$. For the general case of (5), we have

$$\begin{aligned} & \max_{\theta_j} L(\mathbf{r}|\theta_j) \\ &= \max_{\theta_j} \sum_{n=1}^N \log[\mathcal{F}^{-1}\{\Phi_{r_I[n]|H_j}(-\psi_n^I) \Phi_{r_Q[n]|H_j}(-\psi_n^Q)\}], \end{aligned} \quad (10)$$

for $j = 0, 1$, where $\mathcal{F}^{-1}(\cdot)$ denotes inverse Fourier transform. Then, for alternative hypothesis, the maximum of the LF is

$$\begin{aligned} & \max_{\theta_1} L(\mathbf{r}|\theta_1) \\ &= \max_{[\mu_I \ \mu_Q]^T} \sum_{n=1}^N \log[\mathcal{F}^{-1}\{\Phi_{r_I[n]|H_1}(-\psi_n^I) \Phi_{r_Q[n]|H_1}(-\psi_n^Q)\}], \\ & \Phi_{r_I[n]|H_1}(\psi_n^I) = \exp\{j\psi_n^I \mu_I[n] - \gamma|\psi_n^I|^\alpha\}, \\ & \Phi_{r_Q[n]|H_1}(\psi_n^Q) = \exp\{j\psi_n^Q \mu_Q[n] - \gamma|\psi_n^Q|^\alpha\}. \end{aligned} \quad (11)$$

For (11), the MLE of the test vector is $\theta_1^* = [\mu_I^* \ \mu_Q^*]^T$. By inserting the MLE into the (11), we reach

$$\begin{aligned} & \max_{\theta_1} L(\mathbf{r}|\theta_1) \\ &= \sum_{n=1}^N \log[\mathcal{F}^{-1}\{\Phi_{r_I[n]|H_0}(-\psi_n^I) \Phi_{r_Q[n]|H_0}(-\psi_n^Q)\}], \\ & \Phi_{r_I[n]|H_1}(\psi_n^I) = \exp\{j\psi_n^I \mu_I^*[n] - \gamma|\psi_n^I|^\alpha\}, \\ & \Phi_{r_Q[n]|H_1}(\psi_n^Q) = \exp\{j\psi_n^Q \mu_Q^*[n] - \gamma|\psi_n^Q|^\alpha\}. \end{aligned} \quad (12)$$

For the null hypothesis, since the test vector is determined and is single-valued under H_0 , its LF is

$$\begin{aligned} L(\mathbf{r}|\theta_0) &= \sum_{n=1}^N \log[\mathcal{F}^{-1}\{\Phi_{r_I[n]|H_0}(-\psi_n^I) \Phi_{r_Q[n]|H_0}(-\psi_n^Q)\}], \\ \Phi_{r_I[n]|H_0}(\psi_n^I) &= \exp\{-\gamma|\psi_n^I|^\alpha\}, \\ \Phi_{r_Q[n]|H_0}(\psi_n^Q) &= \exp\{-\gamma|\psi_n^Q|^\alpha\}. \end{aligned} \quad (13)$$

By plugging (12) and (13) into (9), the LRT detector can be found for all values of α . However, the derived test statistic has a closed-form formula for only the special cases where the PDF of the received signal is closed-form. For this argument, firstly, the extremely impulsive noise with the Cauchy distribution is considered. By setting $\alpha = 1$, the $S\alpha S$ distribution is reduced to the well-known Cauchy distribution. Due to heavy tails of the Cauchy distribution, abrupt peaks and impulses in the noise samples are more probable than the Gaussian noise. Hence, the Cauchy distribution can adequately describe the statistical behavior of existing impulsive noise in specific environments [23], [40]. The PDF of a real random variable, say x , following the Cauchy distribution ($x \sim \text{Cauchy}(\mu_0, \gamma)$) is

$$f_X(x) = \frac{1}{\pi} \frac{\gamma}{(x - \mu_0)^2 + \gamma^2}. \quad (14)$$

Here, we assumed that I and Q components of the noise are statistically independent while the complex Cauchy noise is not circularly symmetric because of the property of the stable distribution [39]. Now, (6) and (14) for $\alpha = 1$, yield

$$\begin{aligned} r_I[n]|H_0, r_Q[n]|H_0 &\sim \text{Cauchy}(0, \gamma), \\ r_I[n]|H_1 &\sim \text{Cauchy}(\mu_I[n], \gamma), \\ r_Q[n]|H_1 &\sim \text{Cauchy}(\mu_Q[n], \gamma), \end{aligned} \quad (15)$$

and (12) is simplified to

$$\begin{aligned} & \max_{\theta_1} L(\mathbf{r}|\theta_1) \\ &= \sum_{n=1}^N \log\left[\frac{1}{\pi^2} \frac{\gamma}{(r_I[n] - \mu_I^*[n])^2 + \gamma^2} \frac{\gamma}{(r_Q[n] - \mu_Q^*[n])^2 + \gamma^2}\right]. \end{aligned} \quad (16)$$

One can readily show that for the cases of Cauchy and Gaussian distributions the MLEs are $\mu_I^* = \mathbf{r}_I$, $\mu_Q^* = \mathbf{r}_Q$, where $\mathbf{r}_I = [r_I[1] \dots r_I[N]]^T$, and $\mathbf{r}_Q = [r_Q[1] \dots r_Q[N]]^T$. By substituting the MLE of the location parameter, (16) becomes equal to $L(\mathbf{r}|H_1) = N \log[\frac{1}{\pi^2 \gamma^2}]$. Similarly, in the case of the null hypothesis, H_0 , we have

$$L(\mathbf{r}|\theta_1) = \sum_{n=1}^N \log\left[\frac{1}{\pi^2} \frac{\gamma}{(r_I[n])^2 + \gamma^2} \frac{\gamma}{(r_Q[n])^2 + \gamma^2}\right]. \quad (17)$$

Therefore, by using (9), we can write

$$f(\mathbf{r}) = N \log\left[\frac{1}{\pi^2 \gamma^2}\right] - \sum_{n=1}^N \log\left[\frac{1}{\pi^2} \frac{\gamma}{(r_I[n])^2 + \gamma^2} \frac{\gamma}{(r_Q[n])^2 + \gamma^2}\right]. \quad (18)$$

By transferring the constant terms and few algebraic steps, the LRT statistic is

$$f(\mathbf{r}) = \sum_{n=1}^N \log\left(\frac{(r_I[n])^2 + \gamma^2}{(r_Q[n])^2 + \gamma^2}\right) \underset{H_0}{\overset{H_1}{\geq}} \eta. \quad (19)$$

In the rest of the paper, we refer to the detector in (19) by the Cauchy detector (CD). A similar detector for noise following the Cauchy distribution is reported in [25]. The model considered in [25] is circularly symmetric and bi-variate. However, here, CD is derived for the complex signal model of Section II with independent I and Q components without circular symmetry property [39]. CD is frequently employed for different cases of impulsive noise thanks to its acceptable performance under these cases [25], [41].

In a similar way, we can repeat the LRT steps for the special case of $\alpha = 2$ where I and Q components of noise are independent Gaussian processes which jointly form circularly symmetric complex white Gaussian (CSCWG) noise. The characteristic function of $S\alpha S$ distribution and normal distribution become identical, and the variance of normal distribution is related to the dispersion parameter by $\sigma^2 = 2\gamma$ for real random variables. As a result, in this case with several mathematical manipulations, $f(\mathbf{r})$ given in (9) turns into

$$f(\mathbf{r}) = N \log\left[\frac{1}{4\pi\gamma}\right] - \sum_{n=1}^N \log\left[\frac{1}{4\pi\gamma} \exp\left\{-\frac{(r_I[n])^2 + (r_Q[n])^2}{4\gamma}\right\}\right]. \quad (20)$$

By some simplification, the LRT can be found

$$f(\mathbf{r}) = \frac{H_1}{H_0} \|\mathbf{r}\|^2 \geq \eta, \quad (21)$$

where $\|\cdot\|$ is the 2-norm operator. The detector in (21) is energy detector (ED) which is the conventional solution of the LRT to problem (5) with CSCWG noise. Technically, sensing the spectrum by ED is straightforward as it only requires the energy of the received signal. Since the noise in telecommunication systems can be generated due to various phenomena, the statistical behavior of the noise cannot be always accurately described by normal distribution [24]. It is also noteworthy that the performance of ED dramatically degrades by change of α .

Both CD and ED are model-dependent techniques derived for $\alpha = 1$ or $\alpha = 2$. These methods do not perform properly if the noise characteristic exponent changes due to experiencing a new environment. For instance, [24], [42] report different environments in which $S_{\alpha}S$ distribution for noise model with $\alpha = 1.253$ and $\alpha = 1.4329$ are consistent with measurements. As previously shown, for the general case, the LRT detector has no closed-form test statistic, and it is cumbersome to compute it numerically for every special environment with its specific α , and we assume that the SU only uses one detector. Therefore, a detector with generally acceptable performance in different environments with their special α is required.

B. Binary Hypothesis Testing Problem With the Wald and the Rao Tests

In addition to the LRT, which is a solid and conventional method for solving binary hypothesis testing problems, we also continue with two other methods of the Wald and the Rao tests for cases of $\alpha = 1$, and 2 with closed-form PDF to explore other solutions of the problem. The Wald and the Rao tests for the rewritten form of (5) in (29) can be derived through the following formulas, respectively [5], [43], [44]

$$f_w(\mathbf{r}_{aug}) = \frac{H_1}{H_0} (\boldsymbol{\theta}_1^* - \boldsymbol{\theta}_0)^T \mathbf{I}(\boldsymbol{\theta}_1^*) (\boldsymbol{\theta}_1^* - \boldsymbol{\theta}_0) \geq \eta, \quad (22)$$

$$f_r(\mathbf{r}_{aug}) = \frac{\partial L(\mathbf{r}_{aug}|\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \bigg|_{\boldsymbol{\theta}=\boldsymbol{\theta}_0}^T \mathbf{I}^{-1}(\boldsymbol{\theta}_0) \frac{\partial L(\mathbf{r}_{aug}|\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \bigg|_{\boldsymbol{\theta}=\boldsymbol{\theta}_0} \geq \eta, \quad (23)$$

where $\mathbf{r}_{aug}$ is the augmented real-valued of $\mathbf{r}$ (see Appendix), and $L(\mathbf{r}_{aug}|\boldsymbol{\theta})$ is the LF parameterized with the same test vector of the previous part, $\boldsymbol{\theta}$, and $\mathbf{I}(\boldsymbol{\theta})$ is the Fisher Information Matrix (FIM) given by [5]

$$[\mathbf{I}(\boldsymbol{\theta})]_{ij} = -E\left\{\frac{\partial^2 L(\mathbf{r}_{aug}|\boldsymbol{\theta})}{\partial [\boldsymbol{\theta}]_i \partial [\boldsymbol{\theta}]_j}\right\}. \quad (24)$$

The following remark states the Wald and the Rao tests for the special case of the problem with noise distribution of $S_{\alpha}S$ by $\alpha = 1$.

Remark 1: For the binary hypothesis test of (5) with i.i.d noise samples with the complex Cauchy distribution and

independent I and Q components, the Wald test is identical to that of (21), and the Rao test is

$$f(\mathbf{r}) = \sum_{n=1}^N \left[\frac{r_I[n]^2}{(r_I[n]^2 + \gamma^2)^2} + \frac{r_Q[n]^2}{(r_Q[n]^2 + \gamma^2)^2} \right] \frac{H_1}{H_0} \geq \eta. \quad (25)$$

Proof: See Appendix.

The test statistic of (25) is named the Rao test of the Cauchy distribution (RaoC), and it has a unique property due to heavy tails of the Cauchy distribution which will be discussed later in Section IV. Similarly, the Wald and the Rao tests for the case of $\alpha = 2$ is also equal to (21). As observed, the Wald test, the Rao test, and LRT for $\alpha = 2$ and the Wald test for $\alpha = 1$ all reduce to ED in (21), which is the NP test for detection of an unknown random signal with known power, contaminated by CSCWG noise [5], [43]. Interestingly, the second-order moments for stable distribution with $\alpha \neq 2$ do not exist [39]; thus, detectors based on the second-order moments, such as ED, perform inadequately for $\alpha \neq 2$. This is why in [27], [28], detectors based on FLOMs are proposed, which are given by

$$f(\mathbf{r}) = \sum_{n=1}^N |r[n]|^p \frac{H_1}{H_0} \geq \eta, \quad (26)$$

with $0 < p < 2$. In Section IV, we also report the simulation results of the FLOM test with two different amounts of p for the sake of comprehensive comparisons.

C. Binary Classification Problem

In this section, we propose a general spectrum sensing technique based on deep learning. The main advantage of this approach is that, as opposed to the detectors in the previous section, this method is not limited to one model. From a machine learning viewpoint and using its terminology, the SU is supposed to classify the sensed spectrum based on the received signal into two classes. Then, the proposed detector (classifier) labels the sensed spectrum as a white space, H_0 , or a filled channel, H_1 . Thus, the whole process can be regarded as a binary classification problem. A classifier similar to the detector in (8) is a function of the received data (signal). This classifier and its parameters can be mathematically found through a well-known procedure called learning process. In the learning process, the classifier should be trained by a priori provided training dataset, which is a set of signals with their true labels (state of the spectrum).

Due to the large number of samples of the received signal, conventional classifiers are not capable of addressing the problem with enough accuracy. Furthermore, a series of pre-processing steps are needed to transform the contaminated data by the $S_{\alpha}S$ distributed noise into a feature space with fewer dimensions in order to have an effective classification. Thanks to DNNs, all the pre-processing and classification parts can be done for the real-world problems like the general spectrum sensing problem presented in this paper. Hence, a DNN is selected to solve the spectrum sensing problem.

For a DNN, including several layers of neurons with their weights and biases, the process of learning leads to obtain

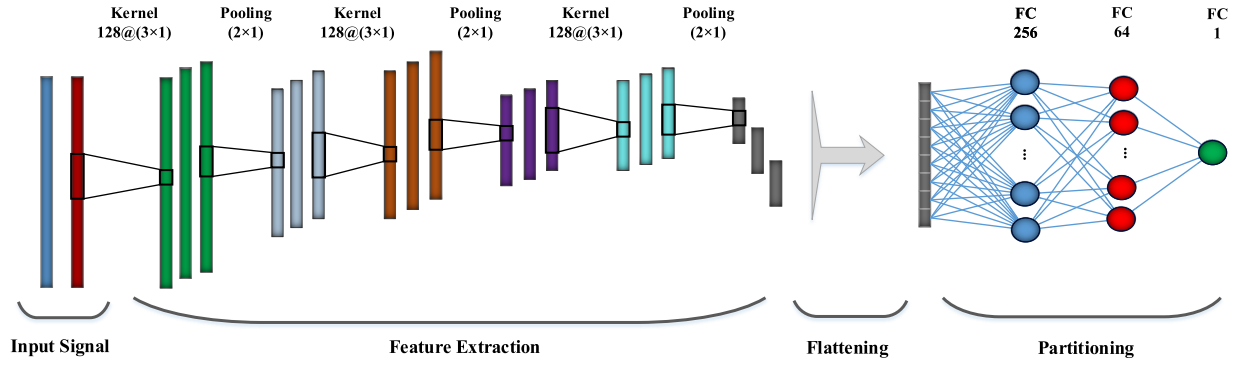


Fig. 1. Convolutional neural network detector with multiple layers of convolution, max pooling, and fully connected.

the appropriate amount of all weights and biases, resulting in favorable positive true and positive false probabilities. To do this, the optimum amount of weights and biases, variables of the optimization problem, can be figured out by minimizing an appropriate cost function. A common and beneficial cost function for binary classification problems is Binary Cross-Entropy (BCE). In the training signal set, $(\mathbf{r}_k, y_k)$, $1 \leq k \leq K$, with K number of signals, $\mathbf{r}_k$ is the k -th training signal belonging to one of the classes (hypotheses). For this problem, just the predicted chance of data belonging to one class is sufficient since the other event is its complement. Thus, the output of the last single neuron, $f(\mathbf{r}_k)$, is used for computing BCE given by:

$$BCE = -\frac{1}{K} \sum_{k=1}^K y_k \log[f(\mathbf{r}_k)] + (1 - y_k) \log[1 - f(\mathbf{r}_k)] \quad (27)$$

where y_k , the label of $\mathbf{r}_k$, is equal to one if $\mathbf{r}_k$ belongs to H_1 ; otherwise, it is zero.

In order to minimize this cost function, the error back-propagation (EBP) algorithm is an adequate solution for the optimization problem in the DNN [14], [15]. Exploiting this algorithm in an iterative manner, for the whole signal set (dataset), each one called one epoch, the network weights approach the optimum weights and biases for the optimization problem.

In this paper, a CNN is employed due to its following advantages for the presented problem. Firstly, an effective fully connected network as deep as a CNN, fed by even a short signal, requires many numbers of wiring and weights, while CNN with shared weights is more efficient. According to the number of input samples, ANNs and SAE with fully connected layers incorporate more weights when the number of input features is large in comparison with CNN. This will happen when a larger N is selected for gaining a higher probability of detection with longer sensing time or the SU senses the spectrum with multiple antennas. However, in Section IV, we will show that CNN outperforms ANN and SAE with the same topologies and the learning algorithms in [30], [38]. Next, in a CNN, max pooling layers extract useful features and ignore infrequent, non-relevant, and redundant patterns. Hence, since the received signal is contaminated by the noise

TABLE I
STRUCTURE OF THE PROPOSED CNN DETECTOR

Type of Layer	Size	Neurons/Filters
convolutional	3×1	128
max pooling	2×1	–
convolutional	3×1	128
max pooling	2×1	–
convolutional	3×1	128
max pooling	2×1	–
FC	–	256
FC	–	64
FC	–	1

process (non-relevant data), a CNN with multiple filters in its feature extracting part, is able to filter the noise and extract the relevant and effective information existing in $s[n]$ for accurate detection of a PU's signal under various noise models. To show this ability of the deep proposed CNN in this paper, we provide the output of the CNN's feature extracting part for both classes in the next section. Topology and size of the kernels and number of neurons of the proposed CNN are listed in Table I. The proposed CNN uses one-dimensional input in spite of CNNs for the image classification. Thus, the original received samples from the sensed channel are inserted to the CNN as a single dimensional array with length of N . However, [35], [37] propose CNNs only practical for multi-antenna SUs, which are inspired by the use of CNNs in image classification. A schematic figure of this proposed CNN detector with single dimensional input is shown in Fig. 1, which provides an insight into the function of the layers and neurons in the proposed CNN.

As shown in Fig. 1, each component of the baseband equivalent complex signal is inputted separately into the network similar to different channels of an RGB image for CNNs working with images. Then, convolution masks are applied to each component, and the outputs of each mask on each component are added together with a bias to form a feature map for the next layer. Except for the last layer, Rectified Linear Unit (ReLU) with $\max\{x, 0\}$ formula is the activation function of all layers to prevent the EBP algorithm from the vanishing gradient. In the final layer, a single neuron with sigmoid function ($\text{Sigmoid}(x) = (1 + \exp(-x))^{-1}$) is used

TABLE II
PROPERTIES OF THE PROPOSED METHODS AND SEVERAL DETECTORS IN THE LITERATURE

Reference	Type	Robust against non-Gaussian Noise	Features/Statistics	Offline Learning Phase	Robust against Parameter's Mismatch	PU's Signal	Number of Antennas
[3],[5]	ED	✗	Energy	✗	✗	General	Single/Extendable
[7]-[9]	SLE	✗	Eigenvalues of SCM	✗	-	General	Only multiple
[7]-[9]	MME	✗	Eigenvalues of SCM	✗	-	General	Only multiple
[8],[9]	AGM	✗	Eigenvalues of SCM	✗	-	General	Only multiple
[8],[9]	CAV	✗	SCM	✗	-	General	Only multiple
[9]	EMR	✗	SCM	✗	-	General	Only multiple
[24]	CCAV	✓	Covariation coefficients	✗	-	General	Only multiple
[25]	COS	✓	See [25]	✗	✗	General	Only multiple
[25], This paper	CD	✓	See (19)	✗	✗	General	Single/Extendable
[29]	FLOM	✓	FLOMS	✗	✗	General	Single/Extendable
This paper	RaoC	✓	See (25)	✗	✗	General	Single/Extendable
[30]	ANN	✗	Energy & Zheng statistic	✓	-	General	Single/Extendable
[31]	CNN	✗	Energy & SCM	✓	-	General	Single/Extendable
[32]	CNN	✗	Energy	✓	-	General	Single/Extendable
[33],[34]	CNN	✗	SCD	✓	-	OFDM	Single/Extendable
[35],[36]	CNN	✗	SCM	✓	-	General	Only multiple
[37]	CNN	✗	SCM	✓	-	General	Only multiple
[38]	SAE	-	Main samples of signal	✓	-	OFDM	Single/Extendable
This paper	CNN	✓	Main samples of signal	✓	✓	General	Single/Extendable

$f(\mathbf{r}) = \text{Sigmoid}(\mathbf{u}^T \mathbf{r}' + b)$, where $\mathbf{u}$ is a vector containing weights to the final neuron, and elements of $\mathbf{r}'$ are outputs of neurons in the previous FC layer, and b is the bias of the last neuron. To have a fixed probability of false alarm, we propose a method to set the probability of the false alarm for the network, P_{fa} , using the bias of the last neuron, b . This method will be discussed in the next section.

IV. SIMULATION RESULTS AND DISCUSSIONS

Here, we enlist several related works in the literature with their properties in Table II. It includes ED, scaled largest eigenvalue (SLE), maximum to minimum eigenvalue (MME), arithmetic-to-geometric mean (AGM), covariance absolute value (CAV), eigenvalue moment ratio (EMR), covariation coefficients absolute value (CCAV), FLOM, CD, and RaoC which are model-based with no offline learning phase. Additionally, for an SU with single antenna, combined with order statistics (COS) detector in [25] for the Gaussian (COS_G) and the Cauchy (COS_C) cases reduce to ED and CD, respectively. Thus, we labeled COS as only for multiple antennas. Moreover, detectors based on various neural networks such as ANN, CNN, and SAE are presented in this table. To the best of our knowledge, the proposed CNN is the first neural network developed to handle non-Gaussian noise. Despite the proposed CNN and SAE, other networks use the energy of samples, SCM, SCD, or a combination of these features as input. These features are related to the second-order moments

of the received signal contaminated by $S\alpha S$ noise, where the second-order moments are not finite. Theoretically, noise variance and covariance matrix are not defined for $S\alpha S$ distribution when $\alpha < 2$. In practice, this phenomenon leads to the unsatisfactory performance of these detectors under this type of noise, and they are sensitive to non-Gaussian noise. To show this phenomenon, we single out ED and ANN with energy statistics in [30] to simulate their performance under $S\alpha S$, but we consider other detectors with these types of features as non-robust detectors against non-Gaussian noise. Furthermore, for detection of i.i.d. signal in Gaussian noise ($\alpha = 2$), ED offers the optimum solution, and its performance is an appropriate benchmark [5], [6]. Additionally, here, we just concentrate on single antenna methods which are consistent with our signal model. For instance, SCM-based detectors require at least two antennas in order to compute SCM. On the other hand, the one-dimensional input of the proposed CNN and SAE helps them to employ the very samples of signal without any preprocessing, whereas other CNN-based detectors require multiple antennas and also a preprocessing to turn the array of signal into a 2-dimensional signal such as SCM. Thus, in the following, we only consider the proposed CNN alongside ED, CD, FLOM, RaoC, ANN, and SAE.

A. Learning Phase of the Proposed CNN Detector

In this part, we assume that $N = 100$, and the receiver is using a single antenna. Here, QPSK modulation is used for

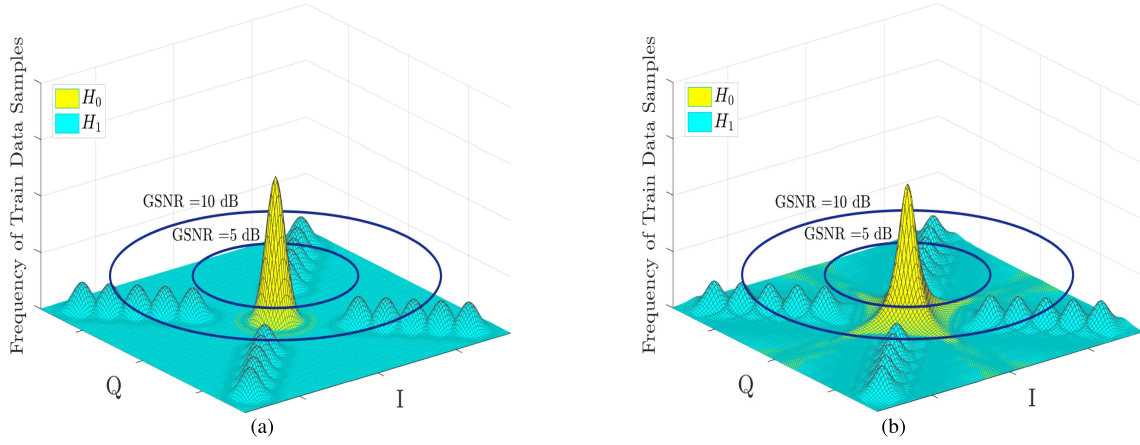


Fig. 2. Statistical frequency of produced signals in 2-dimensional I and Q signal space for QPSK modulation with different noise distribution from the Gaussian distribution with $\alpha = 2$ (a) to extremely impulsive noise of the Cauchy distribution with $\alpha = 1$ (b).

the PU signal. Moreover, we set $\gamma = 0.5$, and the desired false alarm probability for all detectors is $p_{fa} = 0.01$. For the learning process of data-driven detectors, data are produced based on the null and alternative hypotheses. In H_0 , 8×10^4 complex noise signals with $S\alpha S$ distribution, are produced, which all are labeled zero. For each scenario, a specific α from set $\Lambda = \{1, 1.25, 1.5, 1.75, 2\}$ is selected.

For the alternative hypothesis, we assumed that the transmitted signal of the PU comes from a 2-dimensional signal space such as QPSK. To form the signals of H_1 class by different transmitted power, which are contaminated by complex $S\alpha S$ noise with different amounts of α , we need to define a unified parameter for the power of the transmitted signal. Geometric signal to noise ratio [45] and generalized signal to noise ratio (GSNR) in the transmitter [24] are both offered in the literature for this purpose. Here, GSNR is used and defined as $\text{GSNR} = P/\gamma$, where P is the power of $s[n]$. Thus, we have $\|\mathbf{s}\|^2 = NP$, where $\mathbf{s} = [s[1] \dots s[N]]^T$. It is worthwhile to consider the received GSNR in the scenario of the Gaussian noise with $\alpha = 2$. In this scenario, we produced the noise signal in the way that $E\{\|\mathbf{w}\|^2\} = 4N\gamma$, where $\mathbf{w} = [w[1] \dots w[N]]^T$. Now, the received GSNR in the single antenna SU with fading channel and CSCWG noise is denoted by GSNR_r , and is equal to

$$\text{GSNR}_r = \frac{E\{\|\mathbf{h}\mathbf{s}\|^2\}}{E\{\|\mathbf{w}\|^2\}} = \frac{E\{|\mathbf{h}|^2\}E\{\|\mathbf{s}\|^2\}}{4N\gamma} = \frac{P}{4\gamma}, \quad (28)$$

where we used the fact that h and vector $\mathbf{s}$ are independent, and $h \sim \mathcal{CN}(0, 1)$ for all simulations. Therefore, for $\alpha = 2$, $\text{GSNR}_r = \text{GSNR}/4$. For the H_1 hypothesis, 2.5×10^4 complex signals contaminated with the same $S\alpha S$ noise are produced while each one contains a transmitted signal passed through a fading channel with GSNR from -5 dB to 25 dB.

Fig. 2 schematically shows the statistical frequency of signals in the dataset in I and Q 2-dimensional signal space with QPSK modulation. As it can be inferred from Fig. 2, the signals from H_0 class (presented in yellow) are more concentrated around the center point. Contrarily, the statistical frequency of signals belonging to H_1 (presented in cyan) increases around the four symbols of QPSK modulation for

each specific amount of GSNR. For dataset with normally distributed noise ($\alpha = 2$), shown in (a), all the surfaces have light tails, whereas for signals with extremely impulsive noise ($\alpha = 1$), surfaces have heavier tails, shown in (b). In this figure, signals from both classes are more attached together, and they are harder to be separated and classified. As mentioned previously, and it can be observed in this figure, the complex Cauchy distribution is not circularly symmetric since I and Q components are assumed to be statistically independent. In fact, just for the special case of the Gaussian noise in family of the complex stable distributions, circular symmetry and independence of both I and Q can be satisfied simultaneously [39].

For each amount of $\alpha \in \Lambda$, a training dataset is created with totally 10.5×10^4 signals from both classes. The proposed CNN detector is trained for 20 epochs on the dataset. For the proposed SAE by [38], two autoencoders with 100 and 50 neurons are stacked and followed by a logistic regression. ANN in [30] with energy statistics of the current and the previous sensing time is also simulated. The same datasets are employed to train SAE and ANN. Additionally, using the algorithms in [38], training of each autoencoder of SAE includes 500 epochs. By employing EBP algorithm and Adaptive Moment Estimation (Adam) [46], the appropriate weights of the CNN, which minimize the BCE on the dataset, are learned for each scenario in Λ , and they are used for the next simulations. Also, for SAE and ANN, EBP and Adam are utilized more than 20 epochs for the fine-tuning process of SAE network and training ANN network.

B. False Alarm and Threshold Setting

Here, the false alarm probability of detectors versus the amount of the detection threshold is found, and for the proposed CNN detector, the amount of the bias in the single neuron of the last FC layer (for simplicity we refer to it as the last neuron) is adjusted in order to assure that its P_{fa} meets $p_{fa} = 0.01$.

Firstly, the probability of false alarm versus the detection threshold, η , for different amounts of parameter α is simulated.

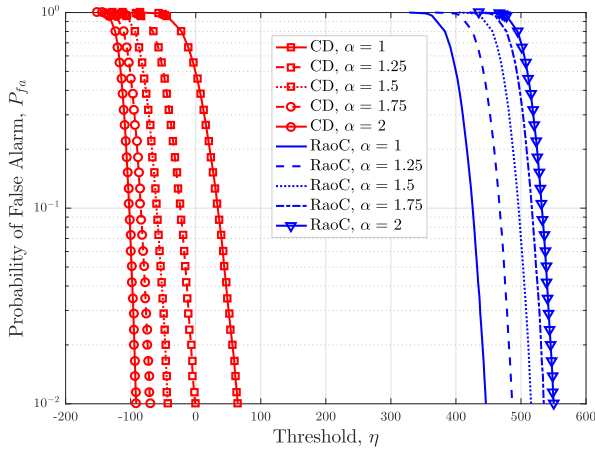


Fig. 3. P_{fa} of CD and RaoC versus detection threshold, η , for $\alpha \in \Lambda$.

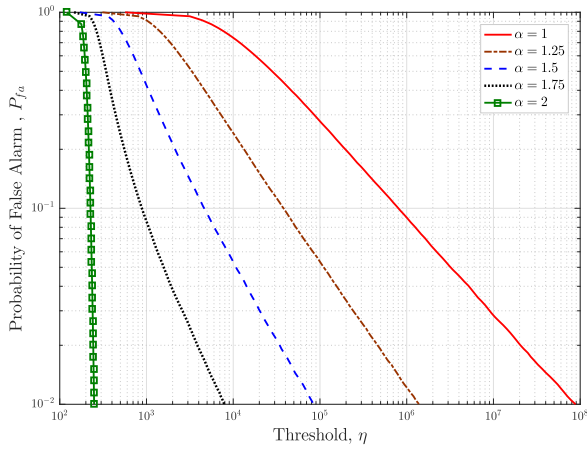


Fig. 4. P_{fa} of ED versus detection threshold, η (logarithmic), for $\alpha \in \Lambda$.

In Fig. 3 the performance of the CD and RaoC are depicted, and their curves are decreasing by growth of η as it was expected. Also, as the samples of the noise process become more impulsive (α decreases), curves of CD shift toward the right side, while curves of RaoC move toward the left side.

Similar curves for FLOM detector with $p = 10^{-2}$ and $p = 10^{-1}$, denoted by FLOM1 and FLOM2, respectively, were simulated while due to limited space they are not presented. As a consequence of the fact that the second-order moments of $S\alpha S$ distribution for $\alpha \neq 2$ tend to infinity, the FLOMs are recommended for detection instead of ED. Fig. 4 includes the same curves for ED. It is worthy to notice that the sharp increase of ED test statistic for signals of class H_0 with $\alpha \neq 2$ leads to large η s. Therefore, for satisfying the constraint $P_{fa} = p_{fa} = 10^{-2}$, the threshold tends to infinity. This phenomenon is verified in Fig 4 when it results in poor performance of ED for $\alpha \neq 2$.

Finally, in this part, we use the bias of the last neuron of CNN for regulating the P_{fa} of the proposed CNN detector while all other weights are set according to the EBP and Adam algorithms. Here, signals from H_0 with specified α are fed into the trained CNN detector for at least $100/p_{fa}$ times, and then for each amount of b , the corresponding P_{fa} is calculated and illustrated in Fig. 5. Thus, based on this

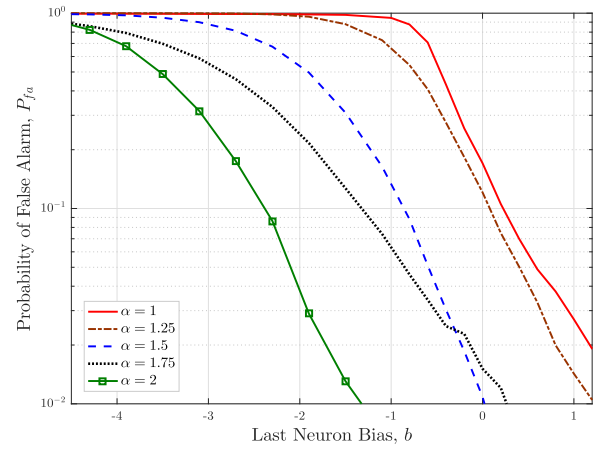


Fig. 5. P_{fa} of the proposed CNN detector for different amounts of b and $\alpha \in \Lambda$.

method, the corresponding bias to $P_{fa} = p_{fa} = 10^{-2}$ is used for the related next simulations that require fixed P_{fa} .

C. Feature Extraction

As shown in Fig.1, the proposed CNN is composed of a feature extracting part followed by FC layers of partitioning part. To investigate the feature extracting part of the proposed CNN, the resulted feature maps (FM) of convolution layers are depicted in Fig. 6. In this figure, (a) is for input with impulsive noise ($\alpha = 1.5$), and (b) is for input with Gaussian noise ($\alpha = 2$). Three cases of input signal for the proposed CNN with absence of PU, presence of PU's weak signal, and presence of PU's strong signal are considered for each subfigure. Thus, corresponding to each of these three scenarios, both (a) and (b) subfigures contain three big blocks. In left blocks, PU's signal is absent, and in the middle and right blocks, PU's signal is present with GSNR of 5 and 20 dB, respectively. For all these three different scenarios, 20 different input signals with $N = 100$ are fed into the proposed CNN. After three layers of convolution and max pooling, FMs with 12 features are extracted for each of 128 filters. All of the 128 FMs are aligned to create a 12×128 strip of FMs for each of those 20 different input signals. Therefore, all these 20 different strips of FMs form a block for each scenario. In left blocks, where PU's signal is absent, there exist lighter blue bars which are barely discernible from the dark blue background especially for (b) with Gaussian noise ($\alpha = 2$). However, these bars are more distinctive in (a) due to spikes of impulsive noise. Middle blocks have more distinctive bars, and those bars have warm colors. Finally, in right blocks with a strong signal of PU, plenty of red bars in each strip exist. Thanks to these red bars, the following FC layers can simply distinguish FM strips of the case with a PU from the case without a PU. Thus, the deep convolutional feature extracting part of CNN can effectively extract suitable features for various noise models.

D. Detection Performance

Now, for examining the detection performance, the produced received signals with various GSNRs are fed into the

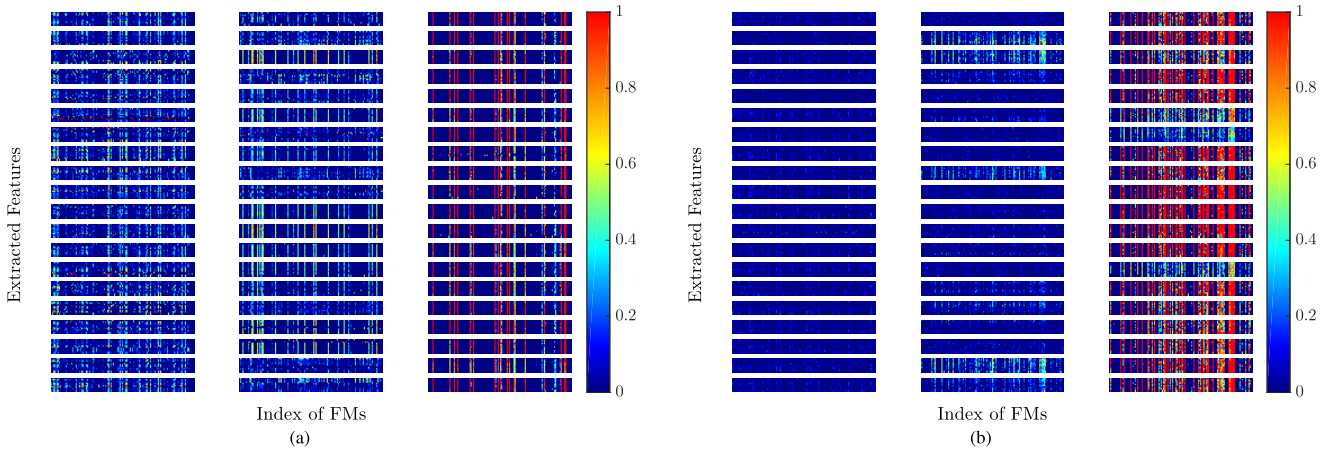


Fig. 6. FMs of different inputs with (a) $\alpha = 1.5$ and (b) $\alpha = 2$. In the left blocks, PU is absent, and in the middle and right blocks, PU's signal with GSNR = 5 and 20 dB exists in sensed channels.

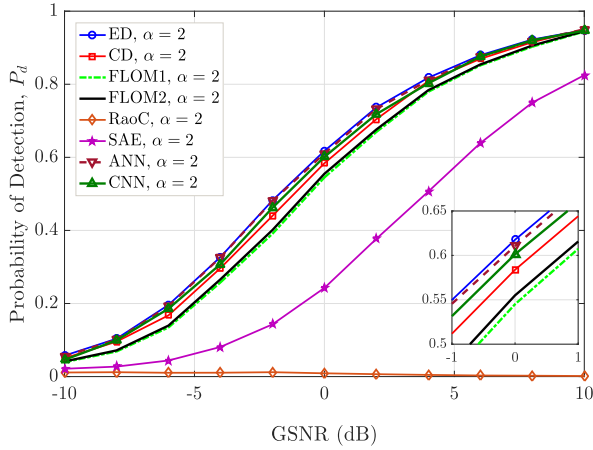


Fig. 7. P_d versus GSNR for the noise model with $\alpha = 2$.

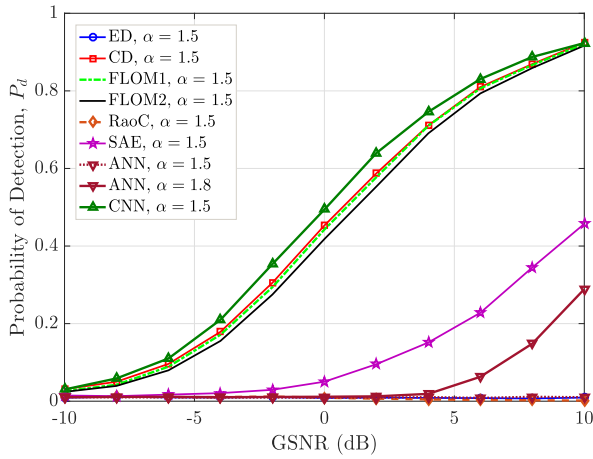
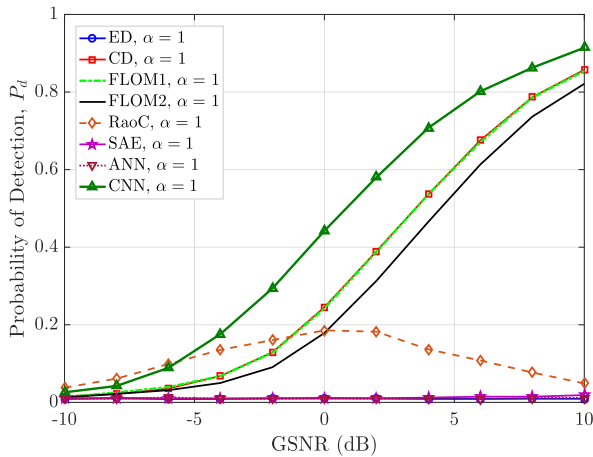
detectors, and then the probability of detection is found. These simulations are repeated for different amounts of α in order to determine the overall performance of all detectors in a variety of physical environments with different statistical behaviors. For more clearance of the figures and curves, three cases of $\alpha = 1, 1.5$, and 2 are just included, and GSNR is limited to the interval of $[-10, 10]$ dB.

Fig. 7 illustrates the probability of detection, P_d , of ED, CD, FLOM1, FLOM2, RaoC, SAE, ANN, and the proposed CNN versus GSNR for the signals contaminated by CSCWG noise with $\alpha = 2$. As expected, ED outperforms other detectors since it is the NP detector for detection of the Gaussian random signals with known power from CSCWG noise [5], and its performance can be considered as a benchmark for comparison. As presented, ANN's performance is roughly identical to ED since it employs energy of samples of the current and the previous sensing intervals as its input features. The proposed CNN's performance is approximately close to benchmark while CD, FLOM2, FLOM1, and SAE have inferior performance, respectively. Also, RaoC performs poorly since it is model-based, and it was derived according to the complex Cauchy distribution.

Similar simulations and curves for the noise signal with complex $S\alpha S$ and $\alpha = 1.5$, which have heavier tails than normal distribution and lighter tails in comparison to the Cauchy distribution, are plotted in Fig. 8. The proposed CNN detector outperforms all other detectors, and for GSNR = 0 dB, it achieves $P_d = 0.5$. FLOMs and CD have smaller detection probability than CNN detector, and performance of ANN and SAE severely degrades under impulsive noise. ANN performs inadequately even for less severe case of impulsive noise with $\alpha = 1.8$.

As was seen in the previous part and Fig. 4, for ED, the detection threshold corresponding to the $p_{fa} = 10^{-2}$ increases when the noise signal becomes more impulsive. For noise distribution with heavy tails, the noise samples with large amplitude are probable, while this chance for normal distribution is almost negligible. Therefore, the η should be sufficiently larger to have the desired false alarm probability. On the other hand, in H_1 , the transmitted signal must have large power, which is related to the location parameter of the received signal distribution in the alternative hypothesis, in order to shift the distribution over that large η for achieving acceptable P_d . Consequently, Fig. 8 confirms that signals with GSNR in this interval are weak, and ED and ANN (with energy statistics as its features) detect them as pure noise. Due to this phenomenon, ED, ANN, and other detectors, which use second-order statistics as input features, are non-robust against impulsive noise.

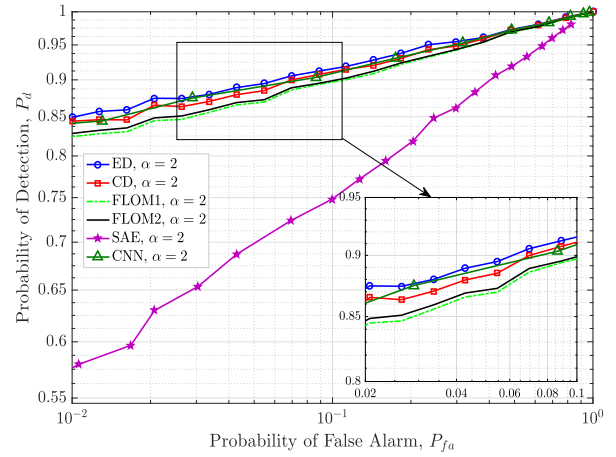
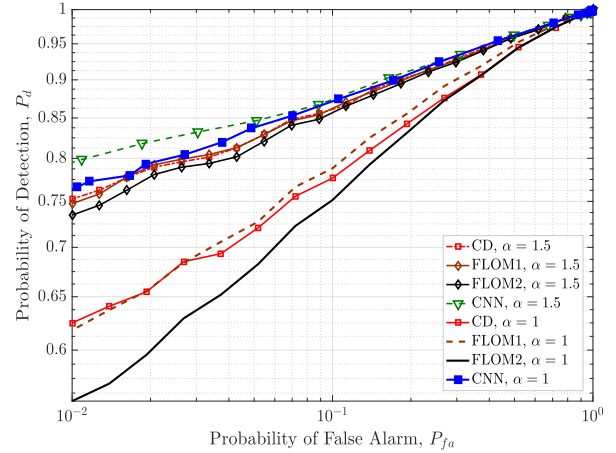
Similarly, the same inference is also applicable for the case of $\alpha = 1$ and the poor detection performance of ED and ANN plotted in Fig. 9. Analogous to previous figures, CNN is more robust to change of noise model. This can be verified by the slight change of P_d when α takes different values. Although CD and RaoC are derived for the case of $\alpha = 1$, they perform poorly in comparison to the proposed CNN detector, especially RaoC. Due to the properties of the Cauchy distribution and its heavy tails, the derived test statistic of RaoC is not an increasing function of GSNR; in fact, it considers the signal samples with high power as pure noise. The same effect can be observed for the locally most powerful (LMP) test of a one-sided hypothesis testing problem with the Cauchy

Fig. 8. P_d versus GSNR for the noise model with $\alpha = 1.5$.Fig. 9. P_d versus GSNR for the noise model with $\alpha = 1$.

noise [4]. Thus, the curve of P_d versus GSNR of RaoC becomes descending for $\text{GSNR} > 1$. Additionally, SAE, FLOM1, and FLOM2 perform worse than the proposed CNN. FLOM2 requires transmitted GSNR of 4.5 dB to obtain $P_d = 0.5$. This value is at least 3 dB more than that of the proposed CNN. At the end of this part, we believe that the proposed CNN offers robustness against impulsive noise in comparison to detectors based on second-order statistics. Also, as shown in the previous subsection, the sufficiently deep architecture of the proposed CNN can extract highly effective features for gaining appropriate detection performance even under the impulsive noise.

E. Receiver Operating Characteristic Curves

To provide a thorough comparison, we also investigate the receiver operating characteristic (ROC) curves which illustrate the probability of detection versus false alarm probability. In Fig. 10, the ROC curves of ED, CD, FLOM1, FLOM2, SAE, and the proposed CNN detector are illustrated for the case of $\alpha = 2$ when $\text{GSNR} = 5$ dB. The probability of detection in descending order belongs to ED, CNN, CD, FLOM2, FLOM1, and SAE. To obtain more clarity in these figures, we only present methods with high performance. Thus, we exclude ANN and RaoC for high GSNR, ED, and SAE for $\alpha \neq 2$. Again, ED can be considered as the NP-based detector, and

Fig. 10. ROC curves for $\alpha = 2$ and $\text{GSNR} = 5$ dB.Fig. 11. ROC curves for $\alpha = 1$, and 1.5 and $\text{GSNR} = 5$ dB.

it performs well while the proposed CNN also performs quite similarly. However, in Fig. 11, for the cases of $\alpha = 1, 1.5$, the CNN detector has the best performance and the largest probability of detection. The advantage of CNN detector is obvious since even its ROC curve for the noise signal with heaviest tails of the case $\alpha = 1$ (presented in blue line with filled squares) is higher than that of all other detectors in the case of $\alpha = 1.5$. It is noteworthy that two classes are more attached and harder to be separated due to heavy tails of noise distribution in the case of $\alpha = 1$. FLOM2 has the worst performance for $\alpha = 1.5$ among all detectors in Fig. 11. Generally, except for CNN, other detectors derived by the conventional methods of detection theory and heuristic tests are unsatisfactory when the noise becomes heavy-tailed. Thus, they are quite sensitive to the noise model in contrast to the robust performance of the proposed CNN detector.

F. Mismatch of Parameters

In realistic scenarios, the true values of parameters in the statistical model of the received signal are unavailable for the SU, while some detectors require knowledge of them for spectrum sensing. Thus, parameters should be estimated accurately in every environment to be employed in spectrum sensing by such detectors. However, the estimated parameters often suffer from mismatch which leads to the degraded performance of these detectors. On the other hand, detection methods

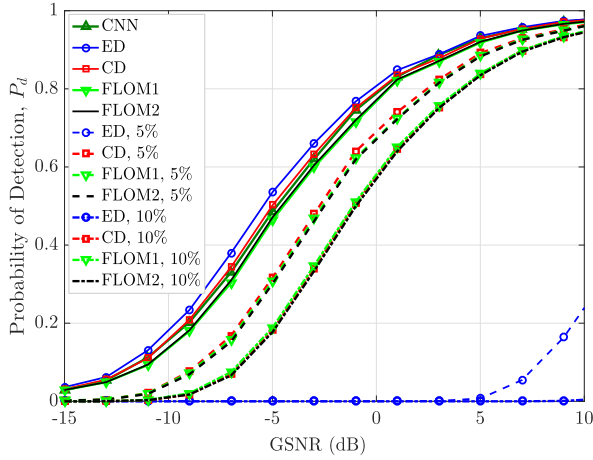


Fig. 12. P_d versus GSNR for $N = 500$ and true value of $\alpha = 2$ with two cases of 5% and 10% estimation error in α .

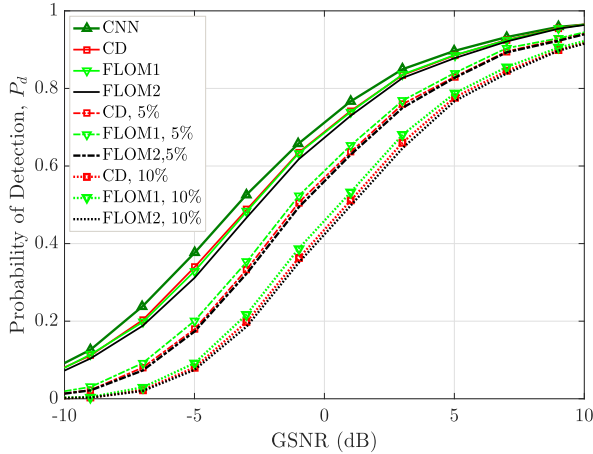


Fig. 13. P_d versus GSNR for $N = 500$ and true value of $\alpha = 1.5$ with two cases of 5% and 10% estimation error in α .

that need no knowledge of parameters of the statistical model are robust against the parameters mismatch. Consequently, data-driven detectors outperform their model-based counterparts under such practical and realistic scenarios.

Here, we investigate the performance of the model-based detectors in comparison with the proposed CNN. In the derivation of model-based detectors, two parameters of α and γ are assumed to be known in the detectors. Therefore, the estimation error in these parameters is troublesome for them. The uncertainty about noise variance and noise power is well-studied in many papers [3], [10], [44]. Thus, our main focus is on the mismatch of α . In the following, we considered two cases of 5% and 10% deviation from the true value of α while it is set to $\alpha = 2$ and 1.5 in Fig. 12 and 13, respectively. It is worthwhile to mention that elongating the sensing time by a detector and gaining more signal samples lead to more accurate detection performance which is favorable in environments with low GSNRs. Therefore, here, we increase N to 500 samples. Also, based on the previous simulation results, curves of detectors with unsatisfactory performance under impulsive noise are excluded. In cases with mismatch of parameters, the detection threshold, η , is adjusted in order

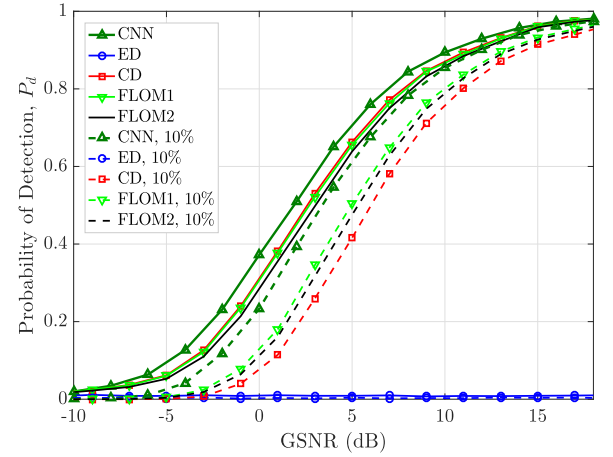


Fig. 14. P_d versus GSNR for $N = 50$ and $(\alpha, \gamma) = (1.5, 0.5)$ with 10% variation in both γ and α of test and train datasets.

to achieve $P_{fa} \leq p_{fa} = 10^{-2}$ (see [10], [44] for more information).

In Fig. 12 with $\alpha = 2$, both CD and the proposed CNN have quite identical results with $N = 500$ which are close to the optimal result of ED. Additionally, under two cases of mismatch, the performance of the proposed CNN is robust thanks to its data-driven property which makes it independent of the amount of α . In contrast to CNN, the performance of FLOMs and CD deteriorate about 2.5 dB and 4 dB for the cases of 5% and 10% error with $P_d = 0.6$, and the most susceptible detector to parameters mismatch is ED. For $\alpha = 1.5$, the best performance between detectors with longer sensing time and robustness against mismatch belongs to the proposed CNN, shown in Fig.13.

For data-driven spectrum sensing approaches, online training or fine-tuning may not be possible in realistic scenarios. Additionally, the SU is required to make a quick decision about the state of the sensed channel in order to shorten the sensing time and enhance its transmission time [9]. Thus, here, we consider the case in which number of sensed samples are reduced to $N = 50$, and both α and γ parameters of the sensed signal (test data) are dissimilar from the train data used for the offline training. In Fig. 14, detection threshold of model-based detectors and weights of the proposed CNN are adjusted based on the produced signal (train data) with $(\alpha, \gamma) = (1.5, 0.5)$. However, the detection probabilities are achieved for the case of identical test and train data and the other case with a 10% discrepancy between (α, γ) parameters of the test and the train data (the test data is generated with $(\alpha, \gamma) = (1.65, 0.45)$ parameters). For having 65% probability of detection, CNN and CD require GSNR of 4 and 4.8 dB while these amounts increase to 5.6 and 8.05 dB for the case with 10% discrepancy. Consequently, the proposed CNN offers the most robust performance among all detectors. Moreover, Fig. 15 corroborates the acceptable performance of the proposed CNN under severe mismatch of parameters of test and train data. In this figure, CNN is trained by employing a train dataset with $(\alpha, \gamma) = (1, 0.5)$. However, the test dataset with 30% mismatch of parameters, $(\alpha, \gamma) = (1.3, 0.35)$, is used to find P_d of detectors for various GSNRs. In Fig. 15, the superiority of the proposed CNN is apparent since its performance, even

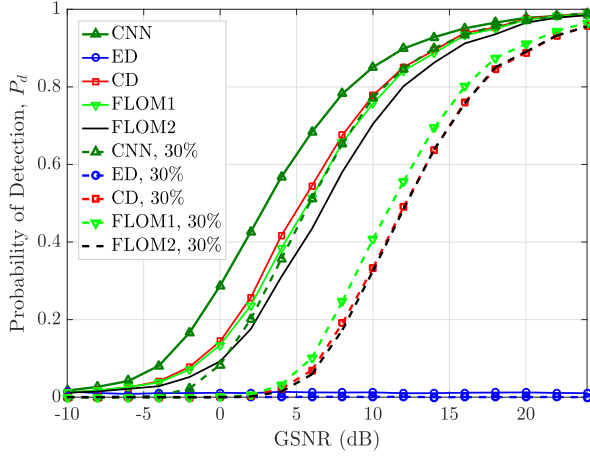


Fig. 15. P_d versus GSNR for $N = 50$ and $(\alpha, \gamma) = (1, 0.5)$ with 30% variation in both γ and α of test and train datasets.

TABLE III
COMPLEXITY OF DETECTORS BASED ON THE
NUMBER OF REAL MULTIPLICATIONS

Detector	Number of Multiplication
ED	$2N$
FLOMs	$2N$
CD	$2N$
RaoC	$2N$
SAE	$2NN_{n_1} + \sum_{l=1}^{L_f-1} N_{n_l}N_{n_{l+1}}$
ANN	$2N + 2N_{n_1} + \sum_{l=1}^{L_f-1} N_{n_l}N_{n_{l+1}}$
CNN	$s_1 N_{f_1} + \sum_{i=1}^{L_c-1} \frac{N}{(p)^i} N_{f_i} K_{s_{i+1}} N_{f_{i+1}} + \frac{N}{(p)^{L_c}} N_{f_{L_c}} N_{n_1} + \sum_{l=1}^{L_f-1} N_{n_l} N_{n_{l+1}}$

under 30% dissimilarity of train and test datasets, is close to the performance of other detectors without dissimilarity.

G. Complexity of Detectors

The computational complexity of detectors is another aspect that we discuss for comprehensive comparisons of these detectors with CNN detector. Here, we consider the number of real multiplications as a criterion for the complexity of detectors, reported in Table III. L_c and L_f denote the number of layers in the feature extracting part and the number of layers in the fully connected part, respectively. Also, N_{f_l} is the number of filters in the l -th convolutional layer, and N_{n_l} is the number of neurons in the l -th fully connected layer. K_{s_l} is the kernel size in the l -th convolutional layer, and p is the size of pooling layers ($p = 2$ in this paper). As seen in Table III, the data-driven detectors require more computational operations, while for the model-based detectors, the order of complexity is N . In contrast to SAE and ANN, which lack convolutional layers, the main burden of the higher complexity for CNN originates from its convolutional layers and its deeper architecture. For the proposed CNN with described architecture in Table I, the hyperparameters are $L_c = 3$, $L_f = 3$, $N_{n_1} = 256$, $N_{n_2} = 64$, and $N_{n_3} = 1$. Also, the kernel size and number of filters are equal to $K_{s_i} = 3$ and $N_{f_i} = 128$ for all convolutional

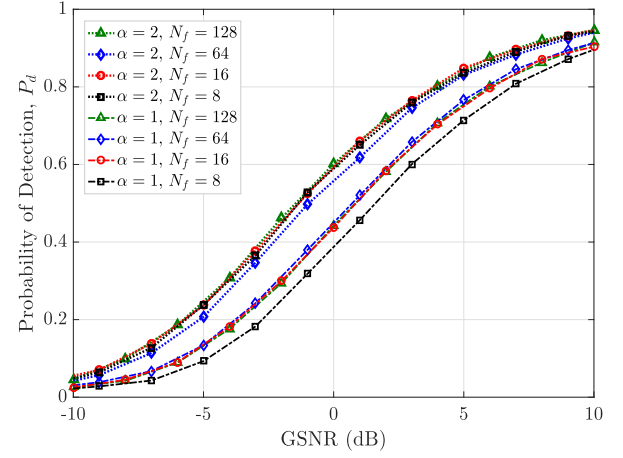


Fig. 16. Probability of detection for CNN versus the GSNR (dB) for the noise signal with $\alpha = 1$ and 2 with various N_f s.

layers, $i = 1, \dots, L_c$. For simulated ANN, we have $L_f = 2$, $N_{n_1} = 7$, and $N_{n_2} = 1$ while parameters of SAE are $L_f = 3$, $N_{n_1} = 100$, $N_{n_2} = 50$, and $N_{n_3} = 1$. In this paper, we mainly used $N = 100$ for the number of received samples. Thus, the number of multiplications with these figures are 221 and 25050 for ANN and SAE respectively. For CNN, it is 4172864 which is greatly larger than other methods.

The process of applying many convolutional filters to feature maps requires a significant number of computational operations. To reduce the computational complexity of the proposed CNN, consequently, we can use fewer number of filters for feature extraction. We assume that a similar number of filters in all convolutional layers are used. This number is denoted by N_f . Now, if N_f decreases from 128 to 64, 16, or 8, the number of multiplications become 1173056, 132800, or 60224, respectively. Thus, the proposed CNN requires more computational operation even by fewer number of filters due to its deeper architecture. However, with $N_f = 8$, the required number of multiplications for SAE and CNN detector have the same order of magnitude. In Fig. 16, we provide the performance of the proposed CNN still with its deep architecture but with fewer number of convolutional filters. As shown, the proposed CNN offers quite identical performance under Gaussian and severe impulsive noise ($\alpha = 1$) even with fewer convolutional filters. Therefore, we conclude that the CNN detector outstrips its rivals due to its deep architecture, and concurrently, its depth imposes a computational burden even with few convolutional filters.

V. CONCLUSION

In this paper, we considered a general noise model encompassing a vast majority of noises in various environments from the common Gaussian noise to severe impulsive noises. This symmetric α -stable model lets us conveniently present the baseband equivalent signal of an SU in any cognitive network. We developed a composite hypothesis testing for the SU which is capable of presenting the spectrum sensing problem in various environments due to the general noise model. We examined the well-known detection techniques such as LRT, Wald, and Rao tests and also machine learning based SAE and ANN detectors for this general framework.

Then, using data-driven neural network, we proposed a novel one-dimensional CNN-based detector. We introduced an effective method to train the CNN for spectrum sensing. Through comparative simulations, we showed the superiority of our method compared to the mentioned detection schemes from different aspects. Our proposed CNN detection method not only outperforms the other detection schemes but it is also more robust against impulsive noise, variations of the noise model, parameters mismatch, and dissimilarity of train and test datasets. This inherent merit is due to the data-driven learning approach exploited in our method. The conventional detection techniques are quite sensitive to these factors. However, the proposed network is more computationally complex in comparison with other investigated detectors.

APPENDIX DERIVATION OF THE WALD AND THE RAO TESTS FOR $\alpha = 1$

For simplicity, firstly, we rewrite (5) into an augmented vector form

$$\begin{cases} H_0 : \mathbf{r}_{aug} = \mathbf{w}_{aug} \\ H_1 : \mathbf{r}_{aug} = \boldsymbol{\mu}_{aug} + \mathbf{w}_{aug}, \end{cases} \quad (29)$$

where $\mathbf{r}_{aug} = [\mathbf{r}_I \ \mathbf{r}_Q]^T$, $\mathbf{w}_{aug} = [\mathbf{w}_I \ \mathbf{w}_Q]^T$, and $\boldsymbol{\mu}_{aug} = [\boldsymbol{\mu}_I \ \boldsymbol{\mu}_Q]^T$, which all are real-valued vectors with a length of $2N \times 1$. Another form of (29) with the same test vector is [5]

$$\begin{cases} H_0 : \boldsymbol{\theta} = \boldsymbol{\theta}_0 \\ H_1 : \boldsymbol{\theta} \neq \boldsymbol{\theta}_0, \end{cases} \quad (30)$$

where we restate $\boldsymbol{\theta} = [\boldsymbol{\mu}_I \ \boldsymbol{\mu}_Q]^T$, $\boldsymbol{\theta}_0 = [\boldsymbol{\mu}_I \ \boldsymbol{\mu}_Q]^T = [\mathbf{0} \ \mathbf{0}]^T$, and $\boldsymbol{\theta}_1 = [\boldsymbol{\mu}_I \ \boldsymbol{\mu}_Q]^T \neq \boldsymbol{\theta}_0$. Thus, under $\alpha = 1$, by use of (16), we have

$$L(\mathbf{r}_{aug}|\boldsymbol{\theta}) = \sum_{n=1}^N \log \left(\frac{\gamma/\pi}{(r_I[n] - \mu_I[n])^2 + \gamma^2} \right) + \sum_{n=1}^N \log \left(\frac{\gamma/\pi}{(r_Q[n] - \mu_Q[n])^2 + \gamma^2} \right). \quad (31)$$

Now, it can be readily shown that

$$[\mathbf{I}(\boldsymbol{\theta})]_{ij} = -E \left\{ \frac{\partial^2 L(\mathbf{r}_{aug}|\boldsymbol{\theta})}{\partial[\boldsymbol{\theta}]_i \partial[\boldsymbol{\theta}]_j} \right\} = 0, \quad i \neq j, \quad (32)$$

and for $i = j$, we have

$$[\mathbf{I}(\boldsymbol{\theta})]_{ii} = -E \left\{ \frac{\partial^2 L(\mathbf{r}_{aug}|\boldsymbol{\theta})}{\partial[\boldsymbol{\theta}]_i^2} \right\} = E \left\{ \left[\frac{\partial L(\mathbf{r}_{aug}|\boldsymbol{\theta})}{\partial[\boldsymbol{\theta}]_i} \right]^2 \right\}. \quad (33)$$

For the second equality, we used the properties of FIM [5], [44]. By using

$$\frac{\partial L(\mathbf{r}_{aug}|\boldsymbol{\theta})}{\partial[\boldsymbol{\theta}]_i} = \begin{cases} \frac{2(r_I[i] - \mu_I[i])}{(r_I[i] - \mu_I[i])^2 + \gamma^2} & i \leq N \\ \frac{2(r_Q[i - N] - \mu_Q[i - N])}{(r_Q[i - N] - \mu_Q[i - N])^2 + \gamma^2} & i > N, \end{cases} \quad (34)$$

and inserting (34) into (33) and using (14), for $i \leq N$, we have

$$[\mathbf{I}(\boldsymbol{\theta})]_{ii} = \frac{4\gamma}{\pi} \int_{-\infty}^{\infty} \frac{(r_I[i] - \mu_I[i])^2}{[(r_I[i] - \mu_I[i])^2 + \gamma^2]^3} dr_I[i]. \quad (35)$$

By this change of variable $y_I[i] = r_I[i] - \mu_I[i]$, and using the fact that the function inside integral is even, (35) becomes

$$[\mathbf{I}(\boldsymbol{\theta})]_{ii} = \frac{8\gamma}{\pi} \int_0^{\infty} \frac{y_I[i]^2}{y_I[i]^2 + \gamma^2} \frac{1}{(y_I[i]^2 + \gamma^2)^2} dy_I[i]. \quad (36)$$

Again, with change of variable $x_I[i] = \frac{1}{y_I[i]^2 + \gamma^2}$ and $y_I[i] = (1/x_I[i] - \gamma^2)^{1/2}$, (36) simplifies to

$$[\mathbf{I}(\boldsymbol{\theta})]_{ii} = \frac{4\gamma}{\pi} \int_0^{1/\gamma^2} x_I[i]^{1/2} (1 - \gamma^2 x_I[i])^{1/2} dx_I[i]. \quad (37)$$

This can be reduced to

$$[\mathbf{I}(\boldsymbol{\theta})]_{ii} = \frac{4}{\pi\gamma^2} \int_0^1 z_I[i]^{1/2} (1 - z_I[i])^{1/2} dz_I[i] \quad (38)$$

by use of $z_I[i] = \gamma^2 x_I[i]$. The integral is in the form of beta function $B(u, v) = \int_0^1 t^{u-1} (1-t)^{v-1} dt = \frac{\Gamma(u)\Gamma(v)}{\Gamma(u+v)}$.

Hence, $[\mathbf{I}(\boldsymbol{\theta})]_{ii} = \frac{4}{\pi\gamma^2} \frac{\Gamma(3/2)^2}{\Gamma(3)} = 1/2\gamma^2$. The same steps can be done for $i > N$, and we reach to the same result. Therefore, the FIM is $2N \times 2N$ diagonal matrix where $\mathbf{I}(\boldsymbol{\theta}) = \frac{1}{2\gamma^2} \mathbf{I}$, and $\mathbf{I}$ is the identity matrix. Now, by putting the FIM, $\boldsymbol{\theta}_1^* = [\boldsymbol{\mu}_I^* \ \boldsymbol{\mu}_Q^*]^T = [\mathbf{r}_I^* \ \mathbf{r}_Q^*]$, and (34), into (22), (23) and few mathematical manipulations, the Wald test in (21) and the RaoC test in (28) can be derived.

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